Junhyeong Kim, Ph.D.

Principal Researcher | 6G Wireless Access System Research Section, ETRI

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Research Profile

Principal Researcher in the 6G Wireless Access System Research Section at ETRI and Assistant Professor in the Major in Information and Communication Engineering at ETRI School, UST. Since joining ETRI in 2011, contributed to 3GPP and IEEE standardization and developed mmWave communication systems for high-speed trains and 5G NR vehicular communications. Research interests include 5G/6G wireless access, wireless standardization, mmWave systems, high-mobility communications, and V2X.

Professional Experience

Principal Researcher — Feb. 2011–Present

6G Wireless Access System Research Section, Electronics and Telecommunications Research Institute (ETRI) — Daejeon 34129, Republic of Korea

- Contributed to 3GPP standardization and IEEE standardization.
- Developed mmWave-band systems for high-speed train communications.
- Developed an mmWave-band 5G NR system for vehicular communications.

Assistant Professor — Sep. 2023–Present

Major in Information and Communication Engineering, ETRI School, University of Science and Technology (UST) — Daejeon 34129, Republic of Korea

Teaching Experience

Advanced Information and Communications System — Spring 2026

Co-Instructor, ETRI School, University of Science and Technology (UST)

Education

Ph.D. in Electrical Engineering — Mar. 2016–Feb. 2020

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Republic of Korea

- Advisor: Prof. Youngnam Han
- Thesis: *Communication Schemes for Platooning in V2X Networks*
- GPA: 3.86/4.3

M.S. in Electrical Engineering — Feb. 2009–Feb. 2011

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Republic of Korea

- Advisor: Prof. Youngnam Han
- Thesis: *Network Coding Based Relay Selection in Two-Source DAS with Interference*
- GPA: 3.82/4.3

B.E. in Electronic Engineering — Aug. 2004–Jul. 2008

Tsinghua University — Beijing, China

- Advisor: Chi Fang
- Thesis: *Static Image Based Face Pose Estimation*
- GPA: 71/100

Selected Project Experience

6G-WISECOM Sep. 2024–Aug. 2027

Development of wireless technology for integrated sensing and communication

- Role: Principal Investigator (PI) and researcher
- Consortium: ETRI (Korea lead), KAIST, Dankook University, University of Oulu (Finland lead)
- Project website: www.6gflagship.com/6g-wisecom

5G-MOBIX Nov. 2019–Oct. 2022

5G for Cooperative & Connected Automated Mobility on X-Border Corridors

- Role: Researcher
- Consortium: 60 partners from 13 countries, including KATECH (Korea lead), ETRI, Ericsson, Nokia, and Ford Otosan
- Project website: www.5g-mobix.com

5G-ALLSTAR Jun. 2018–May 2021

5G AgiLe and fLexible integration of SaTellite And cellulaR

- Role: Researcher (WP2 leader)
- Consortium: ETRI (Korea lead), KT SAT, KATECH, SK Telecom, Snet ICT, CEA-Leti (EU lead), Thales Alenia Space, Grenoble École de Management, Fraunhofer, CRAT
- Project website: 5g-allstar.eu

5GCHAMPION Jun. 2016–May 2019

5G Communication with a Heterogeneous, Agile Mobile network in the Pyeongchang wInter Olympic competitioN

- Role: Researcher
- Project website: www.5g-champion.eu

Mobile Hotspot Network (MHN) Mar. 2012–Feb. 2018

mmWave-band communications for high-speed mobile environments

- Role: Researcher

Honors and Awards

Best Paper Award, ETRI Journal (co-author)	2021
Student Best Paper Award, ISAPE 2021 (co-author)	Dec. 2021
Best Paper Award, IEEE ComComAp 2021 (co-author)	Nov. 2021
Best Paper Award, ETRI Journal (co-author)	2019
Haedong Young Engineer Award	Dec. 2019
Excellent Paper Award, ICTC 2016 (co-author)	Oct. 2016
Best Paper Award, ICTC 2013 (first author)	Oct. 2013

Journal Publications (38)

[1] **J. Kim**, "Outage analysis of platoon-head selection for vehicle platooning in v2x networks," 2026. [Online]. Available: https://arxiv.org/abs/2608.04383

[2] **J. Kim**, Y. Wang, D. Yan, K. Guan, and H. Chung, "Sidelink Evolution Toward Enhanced Data Rate: Experimental and Simulation Validation," *IEEE Transactions on Vehicular Technology*, 2025. [Online]. Available: https://doi.org/10.1109/tvt.2025.3566265

[3] C. Choi, **J. Kim**, and F. Baccelli, "Analysis of a Spatially Correlated Vehicular Network Assisted by Cox-distributed Vehicle Relays," *IEEE Transactions on Vehicular Technology*, 2025. [Online]. Available: https://doi.org/10.1109/tvt.2025.3550181

[4] C. Choi, **J. Kim**, and J. Choi, "Stochastic Geometry Analysis of RIS-Assisted Cellular Networks with Reflective Intelligent Surfaces on Roads," *IEEE Transactions on Communications*, 2024. [Online]. Available: https://doi.org/10.1109/tcomm.2024.3443734

[5] D. Yan, K. Guan, D. He, **J. Kim**, H. Chung, D. Tian, and Z. Zhong, "Blockage effects of road bridge on mmWave channels for intelligent autonomous vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 3, pp. 2908–2919, Mar. 2024. [Online]. Available: https://doi.org/10.1109/TITS.2023.3271133

[6] **J. Kim**, H. Chung, I. Kim, and G. Noh, "Integration of 5G mmWave-enabled V2I and V2V: Experimental evaluation," *IEEE Commun. Mag.*, vol. 62, no. 1, pp. 104–110, Jan. 2024. [Online]. Available: https://doi.org/10.1109/MCOM.001.2300279

[7] D. Yan, Z. Li, K. Guan, D. He, X. Cheng, **J. Kim**, H. Chung, and Z. Zhong, "Modeling and analysis of V2I links for the handover situations at mmWave band," *IEEE Trans. Veh. Technol.*, vol. 72, no. 10, pp. 12 450–12 463, Oct. 2023. [Online]. Available: https://doi.org/10.1109/TVT.2023.3271670

[8] **J. Kim**, Y.-J. Choi, G. Noh, and H. Chung, "On the feasibility of remote driving applications over mmWave 5G vehicular communications: Implementation and demonstration," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 2009–2023, Feb. 2023. [Online]. Available: https://doi.org/10.1109/TVT.2022.3210689

[9] D. He, K. Guan, B. Ai, Z. Zhong, **J. Kim**, H. Chung, and A. Hrovat, "Channel measurement and ray-tracing simulation for 77 GHz automotive radar," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 7, pp. 7746–7756, 2022. [Online]. Available: https://doi.org/10.1109/tits.2022.3208008

[10] G. Noh, **J. Kim**, S. Choi, N. Lee, H. Chung, and I. Kim, "Feasibility Validation of a 5G-Enabled mmWave Vehicular Communication System on a Highway," *IEEE Access*, vol. 9, pp. 36 535–36 546, 2021. [Online]. Available: https://doi.org/10.1109/access.2021.3062907

[11] D. Yan, K. Guan, D. He, B. Ai, Z. Li, **J. Kim**, H. Chung, and Z. Zhong, "Channel characterization for vehicle-to-infrastructure communications in millimeter-wave band," *IEEE Access*, vol. 8, pp. 42 325–42 341, 2020. [Online]. Available: https://doi.org/10.1109/access.2020.2977208

[12] **J. Kim**, G. Casati, N. Cassiau, A. Pietrabissa, A. Giuseppi, D. Yan, E. C. Strinati, M. Thary, D. He, K. Guan, H. Chung, and I. Kim, "Design of cellular, satellite, and integrated systems for 5G and beyond," *ETRI Journal*, vol. 42, no. 5, pp. 669–685, Oct. 2020. [Online]. Available: https://doi.org/10.4218/etrij.2020-0156

[13] E. C. Strinati, S. Barbarossa, T. Choi, A. Pietrabissa, A. Giuseppi, E. D. Santis, J. Vidal, Z. Becvar, T. Haustein, N. Cassiau, F. Costanzo, **J. Kim**, and I. Kim, "6G in the sky: On-demand intelligence at the edge of 3D networks," *ETRI Journal*, vol. 42, no. 5, pp. 643–657, Oct. 2020. [Online]. Available: https://doi.org/10.4218/etrij.2020-0205

[14] L. Ma, K. Guan, D. Yan, D. He, N. Leonor, B. Ai, and **J. Kim**, "Satellite-terrestrial channel characterization in high-speed railway environment at 22.6 GHz," *Radio Science*, vol. 55, no. 3, pp. 1–15, 2020. [Online]. Available: https://doi.org/10.1029/2019rs006995

[15] L. Zhu, D. He, B. Ai, K. Guan, S. Dang, **J. Kim**, H. Chung, and Z. Zhong, "A ray tracing and joint spectrum based clustering and tracking algorithm for internet of intelligent vehicles," *Journal of Communications and Information Networks*, vol. 5, no. 3, pp. 265–281, 2020. [Online]. Available: https://doi.org/10.23919/jcin.2020.9200890

[16] L. Wang, B. Ai, D. He, K. Guan, J. Zhang, **J. Kim**, and Z. Zhong, "Vehicle-to-infrastructure channel characterization in urban environment at 28 GHz," *China Communications*, vol. 16, no. 2, pp. 36–48, 2019. [Online]. Available: https://doi.org/10.12676/j.cc.2019.02.002

[17] H. Yi, K. Guan, D. He, B. Ai, J. Dou, and **J. Kim**, "Characterization for the Vehicle-to-Infrastructure Channel in Urban and Highway Scenarios at the Terahertz Band," *IEEE Access*, vol. 7, pp. 166 984–166 996, 2019. [Online]. Available: https://doi.org/10.1109/access.2019.2953890

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[21] D. Yan, H. Yi, D. He, K. Guan, B. Ai, Z. Zhong, **J. Kim**, and H. Chung, "Channel Characterization for Satellite Link and Terrestrial Link of Vehicular Communication in the mmWave Band," *IEEE Access*, vol. 7, pp. 173 559–173 570, 2019. [Online]. Available: https://doi.org/10.1109/access.2019.2956821

[22] L. Wang, B. Ai, D. He, H. Yi, Z. Zhong, and **J. Kim**, "Channel characterisation in rural railway environment at 28 GHz," *IET Microwaves, Antennas & Propagation*, vol. 13, no. 8, pp. 1052–1059, 2019. [Online]. Available: https://doi.org/10.1049/iet-map.2018.5811

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[28] K. Sakaguchi, T. Haustein, S. Barbarossa, E. C. Strinati, A. Clemente, G. Destino, A. Pärssinen, I. Kim, H. Chung, **J. Kim**, W. Keusgen, R. J. Weiler, K. Takinami, E. Ceci, A. Sadri, L. Xian, A. Maltsev, G. K. Tran, H. Ogawa, K. Mahler, and R. W. H. Jr, "Where, When, and How mmWave is Used in 5G and Beyond," *IEICE Transactions on Electronics*, vol. 100, no. 10, pp. 790–808, Oct. 2017. [Online]. Available: https://doi.org/10.1587/transele.e100.c.790

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Conference Publications (59)

[1] 김진경, 곽병재, 고영조, and **김준형**, “통합 센싱 및 통신 (ISAC) 의 간섭의 대한 개관,” in *한국통신학회 학술대회논문집*, 2025, pp. 470–472.

[2] **J. Kim**, S.-W. Choi, and H. Chung, “Moving Reconfigurable Intelligent Surfaces: A Promising Frontier for 6G Communications,” in *2024 Fifteenth International Conference on Ubiquitous and Future Networks (ICUFN)*, Jul. 2024, pp. 233–237. [Online]. Available: https://doi.org/10.1109/icufn61752.2024.10624985

[3] S.-W. Choi, **J. Kim**, and H. Chung, “Insight Into Ris-Aided Communication Systems Using Path Loss Model,” in *2024 15th International Conference on Information and Communication Technology Convergence (ICTC)*, Oct. 2024, pp. 6061–6063. [Online]. Available: https://doi.org/10.1109/ictc62082.2024.10827176

[4] S.-W. Choi, H. Chung, D.-S. Cho, J.-P. Choi, S.-A. Kim, **J. Kim**, M. Park, N. Lee, J. Song, and N. Sung, “V2I and V2V service demonstration of millimeter wave communication in urban road environment,” in *2023 14th International Conference on Information and Communication Technology Convergence (ICTC)*, Oct. 2023, pp. 756–759. [Online]. Available: https://doi.org/10.1109/ictc58733.2023.10392736

[5] N. Cassiau, I. Kim, E. C. Strinati, G. Noh, A. Pietrabissa, F. Arnal, G. Casati, T. Choi, Y.-J. Choi, H. Chung, S. Colombero, P. D. Zotto, E. D. Santis, J.-B. Doré, A. Giuseppi, J.-M. Houssin, **J. Kim**, M. Laugeois, F. Pigni, X. Popon, L. Raschkowski, M. Thary, and S. H. Won, “5G-ALLSTAR: Beyond 5G Satellite-Terrestrial Multi-Connectivity,” in *2022 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit)*, Jun. 2022, pp. 148–153. [Online]. Available: https://doi.org/10.1109/eucnc/6gsummit54941.2022.9815664

[6] **J. Kim**, D. Yan, K. Guan, D. He, G. Noh, S.-W. Choi, and H. Chung, “Effects of Signal Blockage by a Road Bridge on mmWave Vehicular Communications,” in *2022 13th International Conference on Information and Communication Technology Convergence (ICTC)*, Oct. 2022, pp. 2169–2172. [Online]. Available: https://doi.org/10.1109/ictc55196.2022.9952467

[7] H. Chung, S.-W. Choi, S.-N. Choi, D.-S. Cho, J.-P. Choi, S.-A. Kim, G. Noh, **J. Kim**, M. Park, N. Lee, M. Choi, J. Song, and N. Sung, “Demonstration of millimeter wave vehicle-to-vehicle communication services in highway environment,” in *2022 13th International Conference on Information and Communication Technology Convergence (ICTC)*, Oct. 2022, pp. 1428–1430. [Online]. Available: https://doi.org/10.1109/ictc55196.2022.9952942

[8] S.-W. Choi, **J. Kim**, and H. Chung, “Effective PSCCH Detecting in 5G-NR V2X System,” in *2022 13th International Conference on Information and Communication Technology Convergence (ICTC)*, Oct. 2022, pp. 714–717. [Online]. Available: https://doi.org/10.1109/ictc55196.2022.9952619

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[10] **김준형**, 노고산, and 정희상, “원격 주행 실증을 위한 밀리미터파 5G NR 통신 시스템의 성능 검증,” in *한국통신학회 학술대회논문집*, 2022, pp. 816–817.

[11] H. Chung, **J. Kim**, G. Noh, S. H. Won, T. Choi, and I. Kim, “Demonstration of service continuity based on multi-connectivity with cellular and satellite access networks,” in *2021 International Conference on Information and Communication Technology Convergence (ICTC)*, Oct. 2021, pp. 1400–1402. [Online]. Available: https://doi.org/10.1109/ictc52510.2021.9621102

[12] T. Kim, G. Noh, **J. Kim**, H. Chung, and I. Kim, “Enhanced Resource Allocation Method for 5G V2X Communications,” in *2021 International Conference on Information and Communication Technology Convergence (ICTC)*, Oct. 2021, pp. 621–623. [Online]. Available: https://doi.org/10.1109/ictc52510.2021.9620952

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[15] 정희상, **김준형**, and 노고산, “5G 기반 자율협력주행 모빌리티 실증을 위한 테더링 기술 개발,” in *한국통신학회 학술대회논문집*, 2021, pp. 708–709.

[16] D. Yan, K. Guan, D. He, **J. Kim**, H. Chung, B. Ai, and Z. Zhong, “Channel Characterization for the mmWave Vehicle-to-Vehicle Communication in the See-through situation,” in *2021 13th International Symposium on Antennas, Propagation and EM Theory (ISAPE)*, Dec. 2021, pp. 01–03. [Online]. Available: https://doi.org/10.1109/isape54070.2021.9752939

[17] Y. Song, D. He, D. Yan, K. Guan, Z. Zhong, **J. Kim**, and H. Chung, “Frequency Planning Strategies of Reducing Inter-Cell Interference for MmWave V2I Communication in Urban Scenario,” in *2021 Computing, Communications and IoT Applications (ComComAp)*, 2021, pp. 184–189. [Online]. Available: https://doi.org/10.1109/comcomap53641.2021.9652893

[18] **J. Kim**, G. Casati, A. Pietrabissa, A. Giuseppi, E. C. Strinati, N. Cassiau, G. Noh, H. Chung, I. Kim, M. Thary, J.-M. Houssin, F. Pigni, S. Colombero, P. D. Zotto, L. Raschkowski, and S. Jaeckel, "5G-ALLSTAR: An Integrated Satellite-Cellular System for 5G and Beyond," in *2020 IEEE Wireless Communications and Networking Conference Workshops (WCNCW)*, Apr. 2020, pp. 1–6. [Online]. Available: https://doi.org/10.1109/wcncw48565.2020.9124751

[19] N. Cassiau, G. Noh, S. Jaeckel, L. Raschkowski, J.-M. Houssin, L. Combelles, M. Thary, **J. Kim**, J.-B. Dore, and M. Laugeois, "Satellite and terrestrial multi-connectivity for 5G: Making spectrum sharing possible," in *2020 IEEE Wireless Communications and Networking Conference Workshops (WCNCW)*, Apr. 2020, pp. 1–6. [Online]. Available: https://doi.org/10.1109/wcncw48565.2020.9124728

[20] L. Ma, K. Guan, D. Yan, D. He, B. Ai, **J. Kim**, and H. Chung, "Characterization for High-Speed Railway Channel enabling Smart Rail Mobility at 22.6 GHz," in *2020 IEEE Wireless Communications and Networking Conference (WCNC)*, 2020, pp. 1–6. [Online]. Available: https://doi.org/10.1109/wcnc45663.2020.9120474

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